3. In a $\triangle ABC$, if $\cos C = \dfrac{\sin A}{2\sin B}$, show that the triangle is isosceles.
4. In a $\triangle ABC$, prove that $\dfrac{\sin B}{\sin C} = \dfrac{c - a\cos B}{b - a\cos C}$.
5. In a $\triangle ABC$, prove that $a\cos A + b\cos B + c\cos C = 2a\sin B\sin C$.
6. In a $\triangle ABC$, $\angle A = 60°$. Prove that $b + c = 2a\cos\left(\dfrac{B-C}{2}\right)$.
7. In a $\triangle ABC$, prove the following
   (i) $a\sin\left(\dfrac{A}{2} + B\right) = (b+c)\sin\dfrac{A}{2}$
   (ii) $a(\cos B + \cos C) = 2(b+c)\sin^2\dfrac{A}{2}$
   (iii) $\dfrac{a^2 - c^2}{b^2} = \dfrac{\sin(A-C)}{\sin(A+C)}$
   (iv) $\dfrac{a\sin(B-C)}{b^2 - c^2} = \dfrac{b\sin(C-A)}{c^2 - a^2} = \dfrac{c\sin(A-B)}{a^2 - b^2}$
   (v) $\dfrac{a+b}{a-b} = \tan\left(\dfrac{A+B}{2}\right)\cot\left(\dfrac{A-B}{2}\right)$.
8. In a $\triangle ABC$, prove that $(a^2 - b^2 + c^2)\tan B = (a^2 + b^2 - c^2)\tan C$.
9. An Engineer has to develop a triangular shaped park with a perimeter $120\ m$ in a village. The park to be developed must be of maximum area. Find out the dimensions of the park.
10. A rope of length $12\ m$ is given. Find the largest area of the triangle formed by this rope and find the dimensions of the triangle so formed.
11. Derive Projection formula from (i) Law of sines, (ii) Law of cosines.

3.8 Application to Triangle

Much of architecture and engineering relies on triangular support on any structure where stability is desired. Trigonometry helps to calculate the correct angle for the triangular support. Also trigonometry envisages the builders to correctly layout a curved structure. For a right triangle, any two information with atleast one side say SS, SA are sufficient to find the remaining elements of the triangle. But, to find the solution of an oblique triangle we need three elements with atleast one side. If any three elements with atleast one side of a triangle are given, then the Law of Sines, the Law of Cosines, the Projection formula can be used to find the other three elements.

Working Rule:

- In a right triangle, two sides determine the third side via the Pythagorean theorem and one acute angle determine the other by using the fact that acute angles in a right triangle are complementary.
- If all the sides of a triangle are given, then we can use either cosine formula or half-angle formula to calculate all the angles of the triangle.
- If any two angles and any one of the sides opposite to given angles are given, then we can use sine formula to calculate the other sides.
- If any two sides of a triangle and the included angle are given, we cannot use the Law of sines; but then we can use the law of cosines to calculate other side and other angles of the triangle. In this case we have a unique triangle.
- All methods of solving an oblique triangle require that the length of atleast one side must be provided.